

Solution Requirements

Date	26 June 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in to access the application.	High
2	Authorization levels	Students can upload audio and view reports. Admin/educator can manage users and view all reports.	High

3	External interfaces	Integrates with Speech-to-Text API, NLP models, and PDF report generation libraries.	High
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4	Transactions processing	Accepts audio, converts it to text, analyzes concept understanding, and generates scores and reports.	High
5	Reporting	Generates detailed evaluation reports with transcript, concept score, speech quality, and suggestions.	High
6	Business rules, etc.	Accept only supported audio formats. Calculate scores based on concept similarity, clarity, and speech quality.	High
7	Compliance to laws or regulations	User audio and reports should be handled securely and used only for analysis.	Medium

8	Other	Allow downloading reports and viewing previous analysis results.	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	System should process uploaded audio and generate results in reasonable time.	Audio under 1 minute should be processed within 1–3 minutes.
2	Scalability	Application should be modular so more concepts, users, and reports can be added later.	New modules can be added without changing the full system.
3	Security & Data Privacy	Uploaded audio and generated reports should be handled safely during the session.	Temporary files are deleted after processing.
4	Reliability & Availability	System should handle valid audio files without crashing.	App should show clear errors for unsupported/corrupted audio.

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5	Usability & Accessibility	Interface should be simple for students and educators to use.	Upload, analyze, and download report should be easy to complete.
6	<i>Other</i>	Code should be readable and maintainable.	Separate modules for transcription, scoring, audio features, and report generation.